Interpretable Machine Learning for Predicting the Evolution of the Velocity Gradient Tensor

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Roadmap

- Building on the successful methodology of the **Tensor Basis Neural Network**, we present the existence of a **reduced description** in predicting the mean **deviatoric pressure hessian** using only the Lagrangian **velocity gradient tensor**
- We use this reduced description to **search for interpretability**.

Governing Equations for the Velocity Gradient Tensor

- The **velocity gradient tensor** (VGT) is defined by $A_{ij} = \frac{\partial u_i}{\partial x_j}$, so from Navier Stokes

$$\frac{\partial u_i}{\partial t} + u_k \frac{\partial u_i}{\partial x_k} = -\frac{\partial P}{\partial x_i} + \nu \frac{\partial^2 u_i}{\partial x_k \partial x_k} \tag{1}$$

- We apply spatial derivatives, and use incompressibility to find the governing equations for the **Lagrangian VGT**:

$$\frac{dA_{ij}}{dt} = -\boxed{\left(A_{ik}A_{kj} - \frac{1}{3}A_{mn}A_{nm}\delta_{ij}\right)} - \boxed{\left(\frac{\partial^2 P}{\partial x_i \partial x_j} - \frac{1}{3}\frac{\partial^2 P}{\partial x_k \partial x_k}\delta_{ij}\right)} + \boxed{\nu \frac{\partial^2 A_{ij}}{\partial x_k \partial x_k}} \tag{2}$$

- The main challenge is to predict the nonlocal **deviatoric pressure hessian**, defined as

$$H_{ij} = -\left(\frac{\partial^2 P}{\partial x_i \partial x_j} - \frac{1}{3}\frac{\partial P}{\partial x_k \partial x_k}\delta_{ij}\right) \tag{3}$$

Tensor Basis Neural Network

- We can write the formal solution for the deviatoric pressure hessian as[1]

$$H_{ij}(\mathbf{x}) = \iiint \frac{\delta_{ij} - \hat{r}_i \hat{r}_j}{2\pi r^3} Q(\mathbf{x} + \mathbf{r}) d\mathbf{r} \tag{4}$$

- Using a Taylor expansion, Cayley-Hamilton Theorem, and Tensor Basis expansion[2], we can write

$$\hat{H} = \sum_{i=1}^{10} g^{(i)}(\lambda_1, \ldots, \lambda_5) \cdot \hat{T}^{(i)}(\hat{A}) \tag{5}$$

[1] Ohkitani, K. & Kishiba, S. *Physics of Fluids* 1995.
[2] Pope, S. B. *Journal of Fluid Mechanics* 1975.

Tensor Basis Neural Network

$$\hat{H} = \sum_{i=1}^{10} g^{(i)}(\lambda_1, \ldots, \lambda_5) \cdot \hat{T}^{(i)}(\hat{A}) \tag{6}$$

$$\lambda_1 = \mathrm{tr}(\hat{S}^2) \quad \lambda_2 = \mathrm{tr}(\hat{W}^2) \quad \lambda_3 = \mathrm{tr}(\hat{S}^3) \quad \lambda_4 = \mathrm{tr}(\hat{W}^2\hat{S}) \quad \lambda_5 = \mathrm{tr}(\hat{W}^2\hat{S}^2)$$

$$T^{(1)} = S \qquad T^{(2)} = SW - WS$$

$$T^{(3)} = S^2 - \frac{1}{3} I \cdot \mathrm{tr}(S^2) \qquad T^{(4)} = W^2 - \frac{1}{3} I \cdot \mathrm{tr}(W^2)$$

$$T^{(5)} = WS^2 - S^2 W \qquad T^{(6)} = W^2 S + SW^2 - \frac{2}{3} I \cdot \mathrm{tr}(SW^2)$$

$$T^{(7)} = WSW^2 - W^2 SW \qquad T^{(8)} = SWS^2 - S^2 WS$$

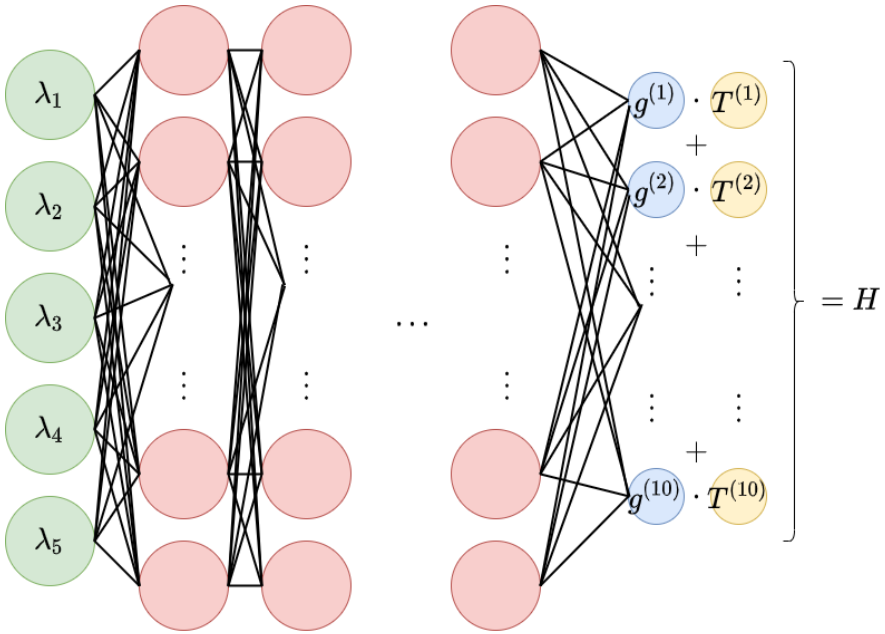
$$T^{(9)} = W^2 S^2 + S^2 W^2 - \frac{2}{3} I \cdot \mathrm{tr}(S^2 W^2) \qquad T^{(10)} = WS^2 W^2 - W^2 S^2 W$$

- We use the natural timescale $\tau = \langle \left\| S^2 \right\|_2 \rangle^{-1}$ to normalize our VGT, and thus all of the λ_i, $T^{(j)}$.

Tensor Basis Neural Network

$$\hat{H} = \sum_{i=1}^{10} g_\theta{}^{(i)}(\lambda_1, \ldots, \lambda_5) \cdot \hat{T}^{(i)}(\hat{A}) \tag{7}$$

$$\lambda_1 = \mathrm{tr}(\hat{S}^2) \quad \lambda_2 = \mathrm{tr}(\hat{W}^2) \quad \lambda_3 = \mathrm{tr}(\hat{S}^3) \quad \lambda_4 = \mathrm{tr}(\hat{W}^2\hat{S}) \quad \lambda_5 = \mathrm{tr}(\hat{W}^2\hat{S}^2)$$



$$L(\theta) = \frac{1}{N} \sum^{N} \left\| \hat{H}_{\text{truth}} - \sum_{i=1}^{10} g_\theta^{(i)}(\lambda_1, \ldots, \lambda_5) T^{(i)} \right\|_2^2 \tag{8}$$

Previous Work

- Previous phenomenological models attempted to use a truncated tensor basis expansion[3,4]
- Tian, et.al[5] showed a significant improvement using the full tensor basis, and learning the $g^{(i)}$ from data.
- **We investigate Tian, et.al's methodology to find that** $g(\lambda_1, \ldots, \lambda_5) \approx g(\lambda_1, \lambda_2)$.

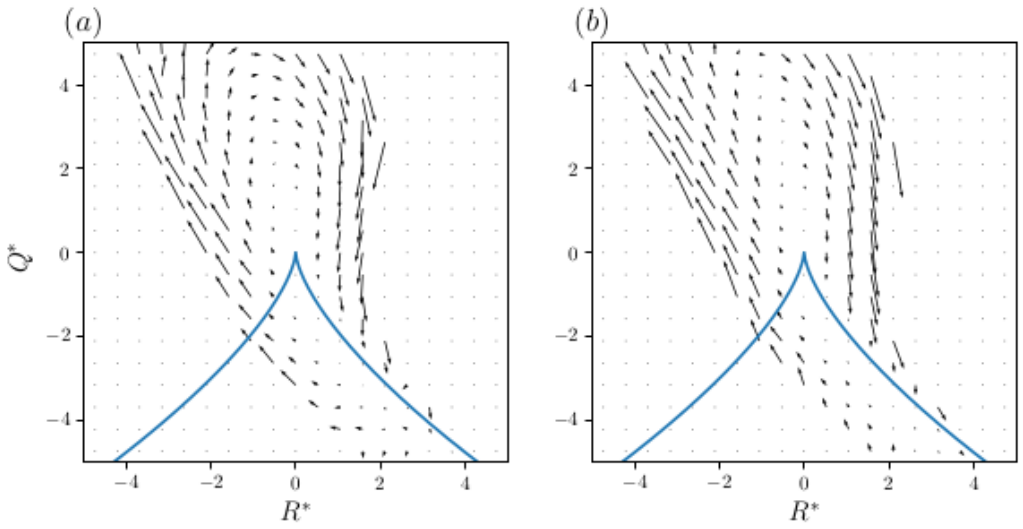


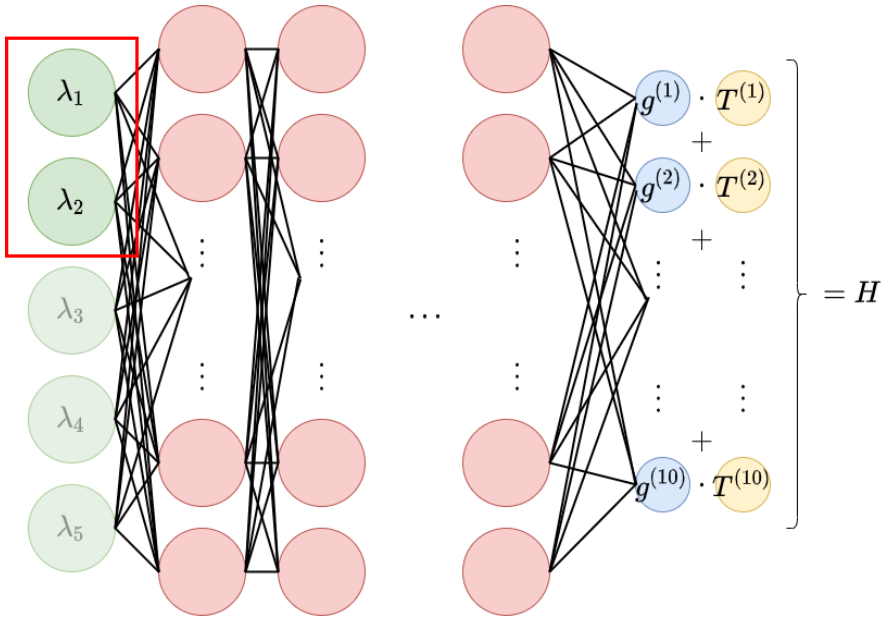
Figure: Results from Tian et.al showing: (left) DNS projection into the QR plane, (right) predicted QR CMTs from the TBNN

[3] Johnson, P. L. & Meneveau, C. *Journal of Fluid Mechanics* 2016.
[4] Lawson, J. M. & Dawson, J. R. *Journal of Fluid Mechanics* 2015.
[5] Tian, Y. *et al.* *Physical Review Fluids* 2021.

Tensor Basis Neural Network

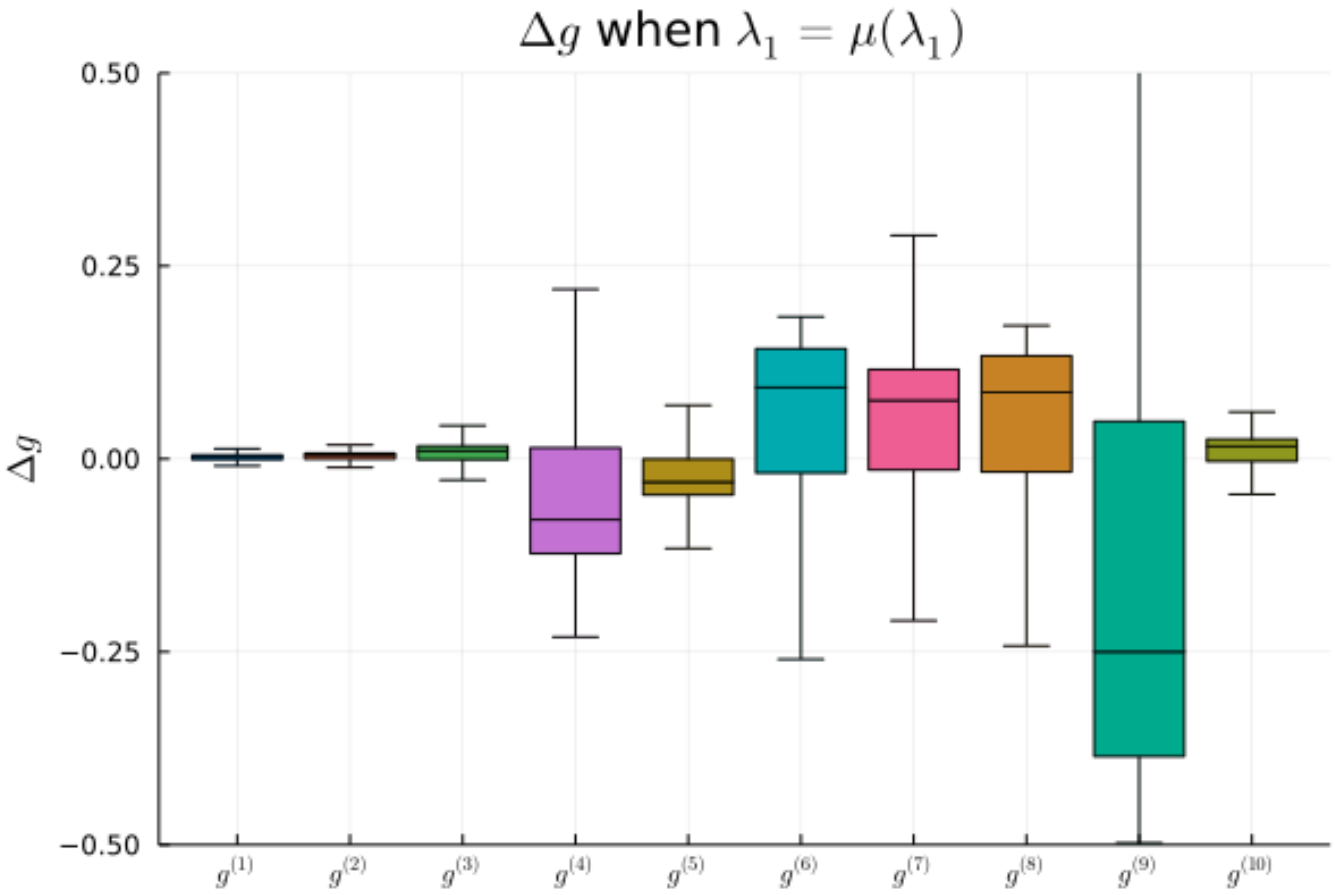
$$\hat{H} = \sum_{i=1}^{10} g_\theta{}^{(i)}(\lambda_1, \ldots, \lambda_5) \cdot \hat{T}^{(i)}(\hat{A}) \tag{9}$$

$$\lambda_1 = \mathrm{tr}(\hat{S}^2) \quad \lambda_2 = \mathrm{tr}(\hat{W}^2) \quad \lambda_3 = \mathrm{tr}(\hat{S}^3) \quad \lambda_4 = \mathrm{tr}(\hat{W}^2\hat{S}) \quad \lambda_5 = \mathrm{tr}(\hat{W}^2\hat{S}^2)$$



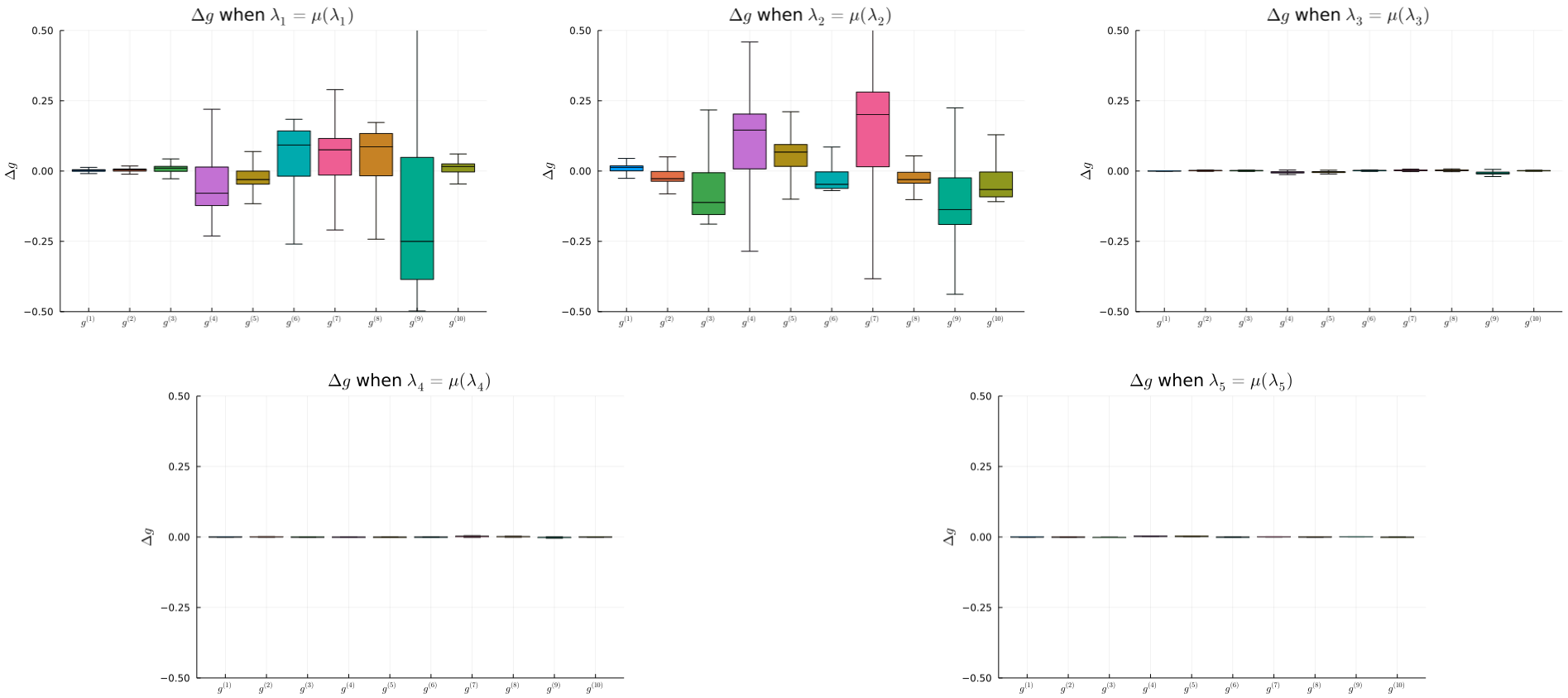
$$L(\theta) = \frac{1}{N} \sum^{N} \left\| \hat{H}_{\text{truth}} - \sum_{i=1}^{10} g_\theta^{(i)}(\lambda_1, \ldots, \lambda_5) T^{(i)} \right\|_2^2 \tag{10}$$

Distributions of sensitivity to parameter variability



- We found the **reduced description** via observing **distributions of sensitivity** of the **scalar functions** $g^{(i)}$ to **variability of parameters** λ_j.

Reduced Description



- We found the **reduced description** via observing **distributions of sensitivity** of the **scalar functions** $g^{(i)}$ to **variability of parameters** λ_j.

A priori tests of reduced description

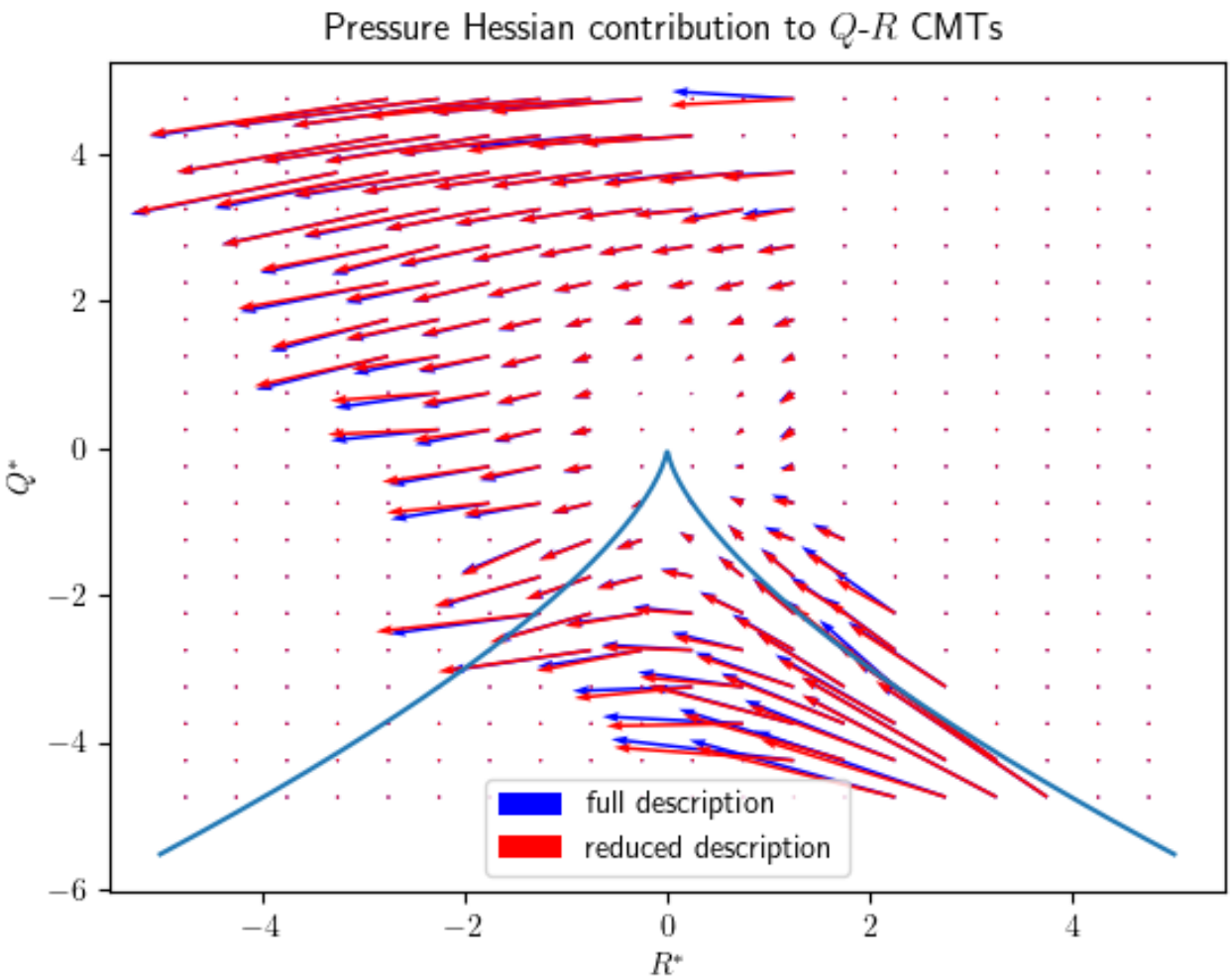
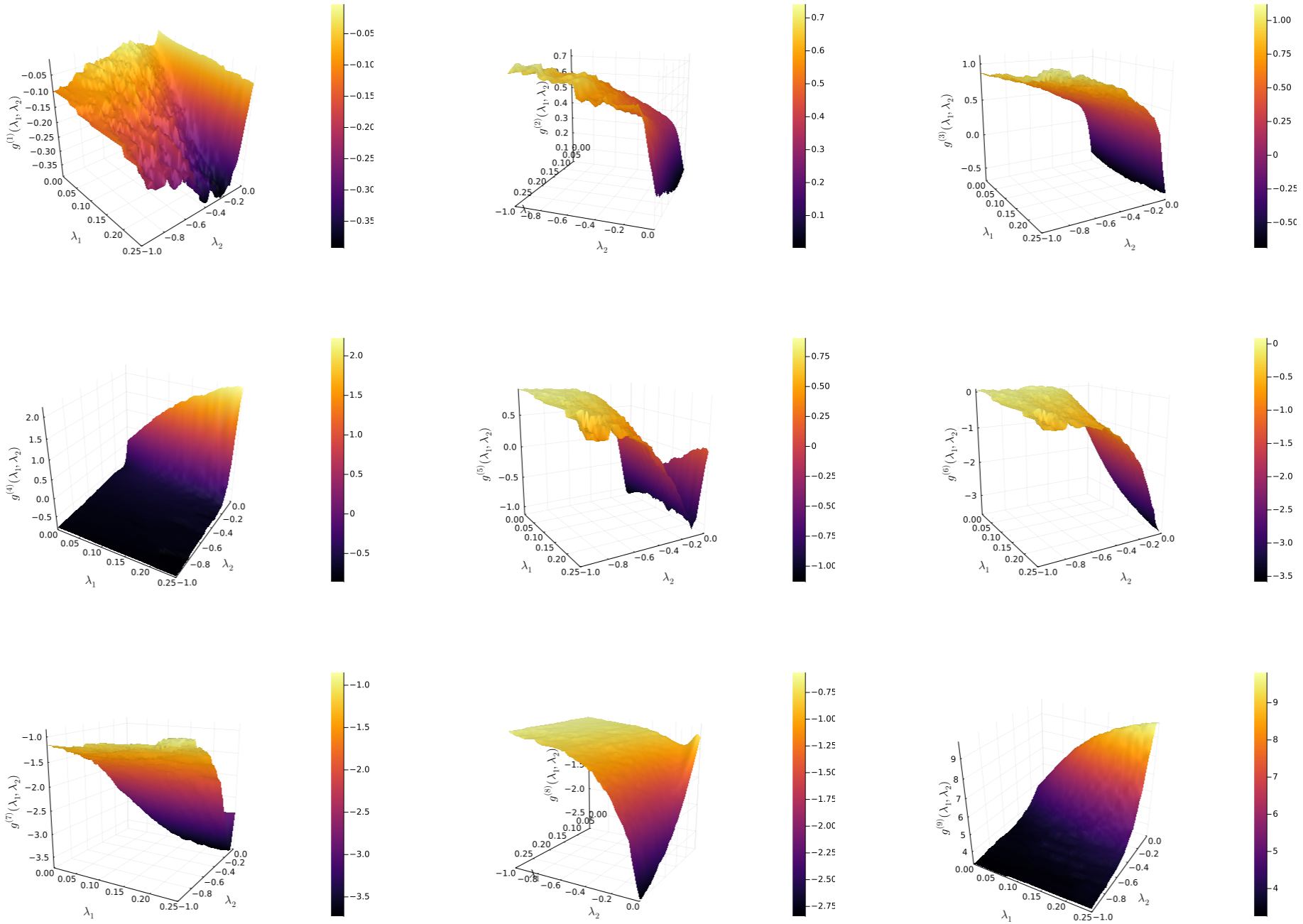
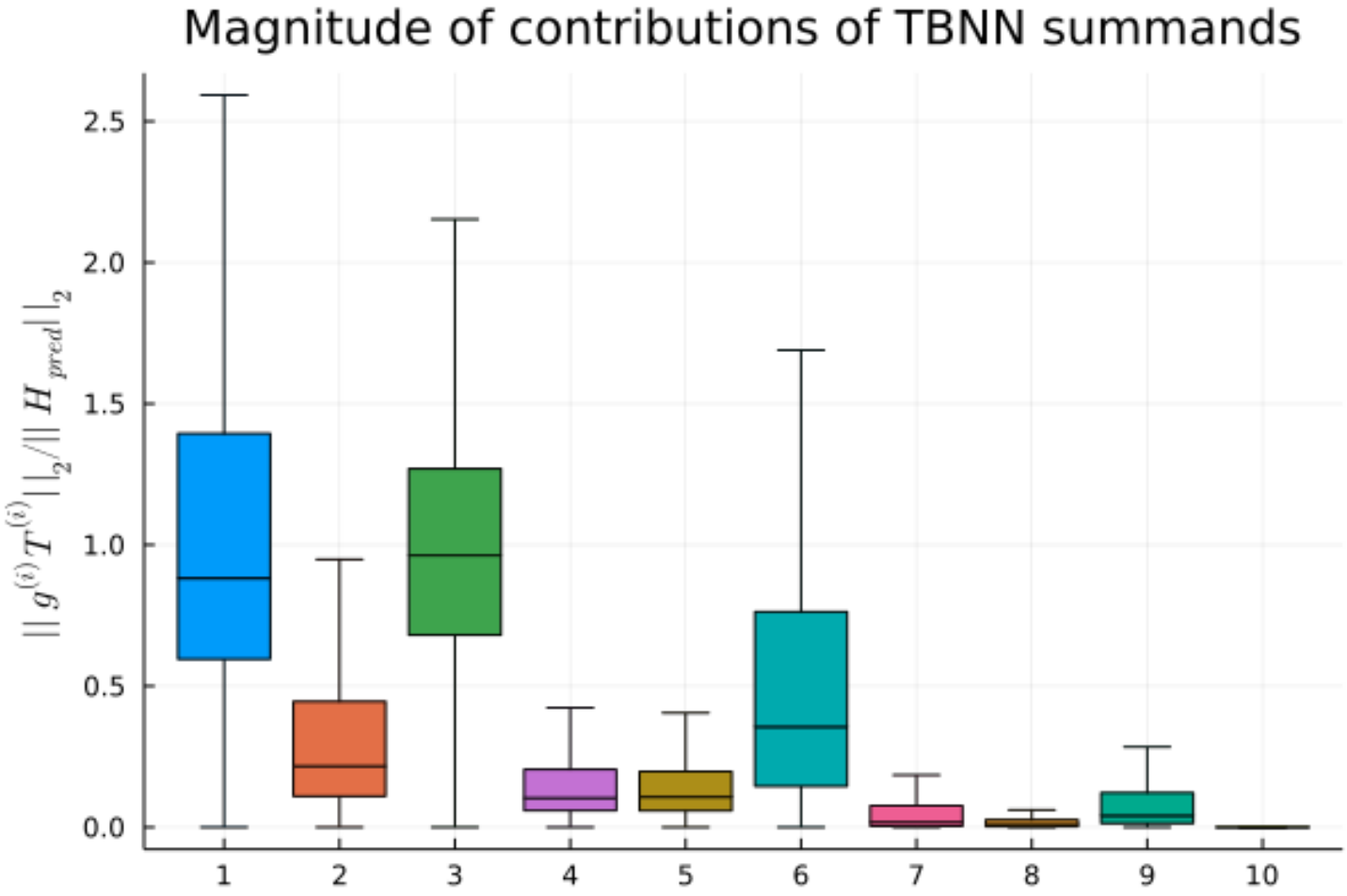


Figure: Comparison of Pressure Hessian contribution to Q-R CMTs using all five invariants (blue), and only the first two (red)

$g^{(i)}(\lambda_1, \lambda_2)$



Is a further reduction possible?



Conclusions & Future Work

- Conclusions
  - We investigated the successful methodology of the TBNN to find there exists a reduced parameterization of the scalar functions weighting the tensor basis contributions.
  - This reduced parameterization allowed us to more readily understand the functional dependence on the invariants
  - This model reduction will be important as we extend the methodology to coarse-grained predictions
- Future Work
  - Does this reduced parameterization generalize over Reynold's number?
  - Can we tease out a functional form for the $g^{(i)}$?

References

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Johnson, P. L. & Meneveau, C. A closure for Lagrangian velocity gradient evolution in turbulence using recent-deformation mapping of initially Gaussian fields. *Journal of Fluid Mechanics* **804,** 387–419 (2016).

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Tian, Y., Livescu, D. & Chertkov, M. Physics-informed machine learning of the Lagrangian dynamics of velocity gradient tensor. *Physical Review Fluids* **6,** 094607 (2021).